

6.5 Homepage of e-Calculus

www.math.uakron.edu/%7Edpstory/e-calculus.html

As in one other site maintained by Prof D P Story that we've mentioned here, this one, too, is nothing site-wise: it's a collection of PDFs. You'll want to download them: they teach the subject well! Here's an extract from somewhere near the beginning of the course:



Prof Story (again), this time on calculus

“Assuming that you are not familiar with the technical aspects of this section, when you think about it, your knowledge of velocity is limited. In terms of your own mathematical background, there is only one type of velocity you can deal with: Constant Velocity. To make matters worse, there is really only one formula for dealing with velocity: the famous formula,

distance = velocity x time

“But the good news is that simple algebra is sufficient to solve many problems involving constant velocity. The bad news is that, based on life’s experiences, velocity is hardly ever constant! One way around this problem is through the notion of average velocity. Even though the average velocity concept is very useful in many different situations, it is inadequate for a deeper study of the dynamics of motion of a particle. For this reason, we need a better understanding of velocity.

“This requires us to move to a higher mathematical plane: The Calculus level. It was the limit concept that enabled mathematicians to move from the algebraic level to the Calculus level.”